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EnerVision: AI-Powered Digital Twin for Predictive Energy Intelligence and Sustainable Manufacturing

An Interdisciplinary Project Report (XX367P)

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In partial fulfillment of the requirements for the degree of

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RV College of Engineering[®], Bengaluru
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DECLARATION

We, **Aaditey Chalva, Ishita Goyal, Manoj.R, Megha Thakur** students of sixth semester B.E., RV College of Engineering, Bengaluru, hereby declare that the interdisciplinary project titled '**EnerVision: AI-Powered Digital Twin for Predictive Energy Intelligence and Sustainable Manufacturing**' has been carried out by us and submitted in partial fulfilment for the award of degree of **Bachelor of Engineering** in respective departments during the year 2024-25.

Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any degree or diploma to any other university.

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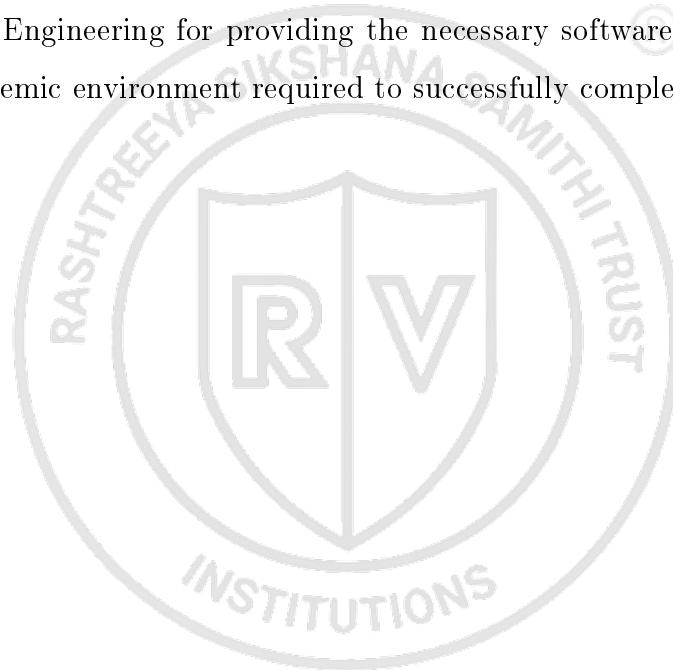
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ABSTRACT

Micro, Small and Medium Enterprises are important contributors to industrial development, but many MSMEs face high energy consumption, high electricity cost, diesel-generator dependency and limited access to intelligent energy management systems. Traditional energy monitoring systems mainly display past or present readings and do not provide prediction, optimization or decision support. Because of this, industries may face energy wastage, peak-load penalties and unnecessary carbon dioxide emissions.

This project presents an AI-driven digital twin for smart energy optimization in MSMEs using IoT-based energy analytics. The system uses historical industrial energy data consisting of production output, grid energy and diesel-generator energy. The data is cleaned and processed using MATLAB, and important Key Performance Indicators such as total energy consumption, Energy Per Unit, energy efficiency, operating cost, generator dependency and carbon dioxide emissions are calculated.

Linear and quadratic regression models are developed to study the relationship between production and total energy consumption. The quadratic and linear regression models are selected as the optimal modeling approach because they provide stable coefficients of determination without overfitting the limited dataset. While complex machine learning techniques such as Random Forest, XGBoost and LSTM forecasting were evaluated, they were excluded from the final digital twin due to severe overfitting and test-leakage caused by the restricted sample size (59 records).

A digital twin simulation engine is developed to test different operational scenarios without disturbing the real manufacturing system. The scenarios include mathematical energy baseline bounding, diesel-generator reduction, production scaling, base-load reduction and peak-load avoidance. A dashboard-based decision support system is proposed to display KPIs, energy trends, anomaly alerts, load scheduling recommendations and what-if simulation results.

The results show that the AI-driven digital twin can support MSMEs in reducing energy cost, improving energy efficiency, decreasing diesel-generator usage and supporting sustainable industrial operation. The project demonstrates how Industry 4.0 technologies such as IoT, AI and digital twins can be used as a low-cost and scalable solution for smart energy management in MSMEs.

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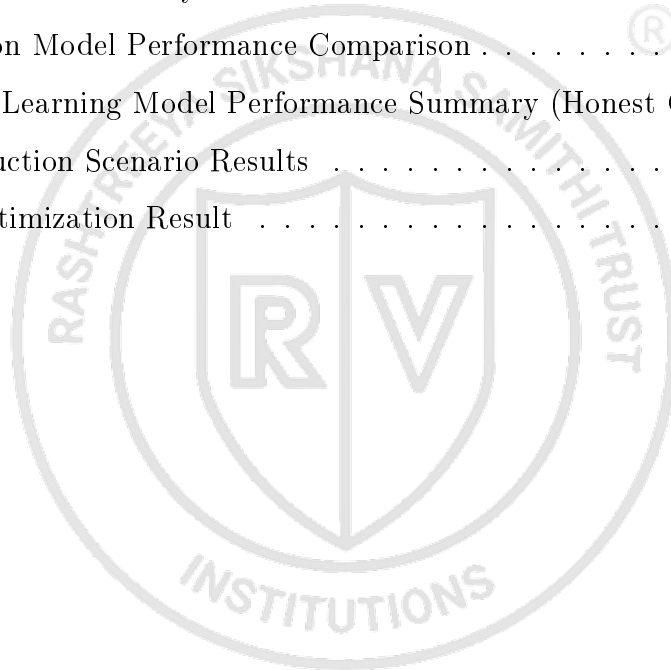
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CHAPTER 1

INTRODUCTION TO AI-DRIVEN DIGITAL TWIN FOR MSME ENERGY OPTIMIZATION

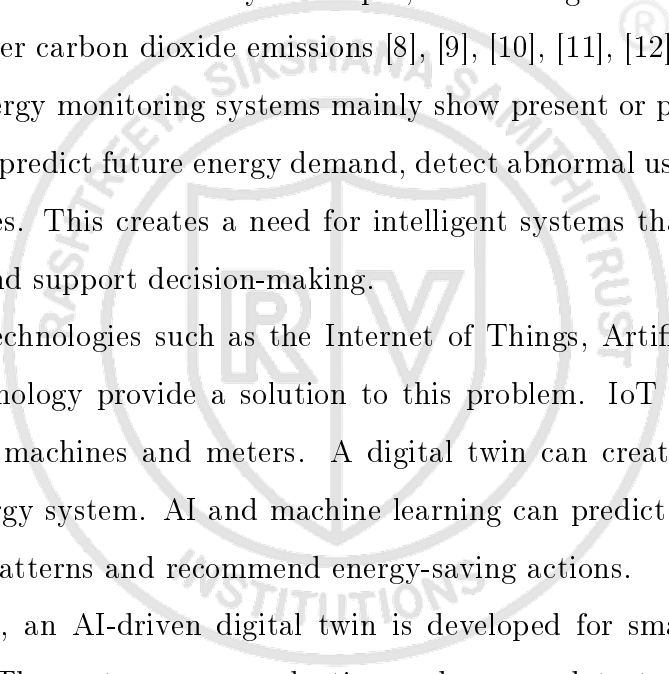
1.1 Introduction

Industrial energy management has become an important part of modern manufacturing because energy cost directly affects production cost and profit. Micro, Small and Medium Enterprises often operate with limited resources, older machinery and basic monitoring systems [1], [2], [3], [4], [5], [6], [7]. Many MSMEs depend on both grid electricity and diesel generators. Grid electricity is cheaper, but diesel-generator power is expensive and produces higher carbon dioxide emissions [8], [9], [10], [11], [12].

Traditional energy monitoring systems mainly show present or past energy consumption. They do not predict future energy demand, detect abnormal usage or suggest better operating schedules. This creates a need for intelligent systems that can collect energy data, analyze it and support decision-making.

Industry 4.0 technologies such as the Internet of Things, Artificial Intelligence and Digital Twin technology provide a solution to this problem. IoT can collect real-time energy data from machines and meters. A digital twin can create a virtual model of the industrial energy system. AI and machine learning can predict future consumption, detect abnormal patterns and recommend energy-saving actions.

In this project, an AI-driven digital twin is developed for smart energy optimization in MSMEs. The system uses production and energy data to predict total energy consumption, analyze generator usage, calculate cost and emissions, and simulate different energy-saving scenarios. The aim is to help MSMEs reduce energy wastage, lower operating cost and improve sustainability.



Production	Predicted Energy	Observation
3500 units	8214.74 kWh	Energy requirement is moderate at lower production.
4500 units	9373.36 kWh	Energy consumption increases as production increases.
5500 units	9942.97 kWh	Highest production needs the highest energy, but energy rise is gradual.

Figure 1.1: Industry 4.0 Based Energy Management Concept

1.2 Literature Review

Digital twin technology has been widely studied for smart manufacturing and industrial energy management. Earlier studies explain how digital twins create virtual models of real systems and help in monitoring, simulation and optimization. Tao et al. discussed the use of digital twins in industry and explained their importance in smart manufacturing. Kritzinger et al. reviewed digital twin applications in manufacturing and described different types of digital twin models.

Research on energy-efficient digital twins shows that virtual models can be used to analyze production behavior and energy consumption under different operating conditions. Studies on IoT-based energy monitoring explain how sensors, smart meters and cloud dashboards can help industries monitor voltage, current, power and energy consumption.

Machine learning has also been used for energy prediction. Regression models, Random Forest, XGBoost and deep learning models such as LSTM can predict future energy demand from historical data. Anomaly detection methods can identify abnormal energy usage caused by machine faults, sudden load increase or inefficient operation.

However, many existing systems focus only on one part of the problem. Some systems only monitor energy, while others only predict demand. Many studies are designed for large industries and may not be suitable for MSMEs. There is a need for a low-cost integrated framework that combines data collection, KPI analysis, prediction, scenario simulation, anomaly detection and dashboard-based decision support.

1.2.1 Summary of Literature Gap

1. Most digital twin systems are designed for large industries and are not cost-effective for MSMEs.
2. Many systems provide monitoring but do not include prediction and optimization.
3. Existing studies often focus on a single technique such as regression, ML prediction or anomaly detection.
4. Few systems combine energy KPIs, machine learning, generator-usage prediction, scenario simulation and dashboard support in one platform.
5. MSMEs require a simple, low-cost and interpretable solution for practical energy management.

6. This project addresses these gaps by developing an integrated AI-driven digital twin framework for MSME energy optimization.

1.3 Motivation

The motivation for this project comes from the need to reduce energy wastage and operating cost in MSMEs. Many small and medium manufacturing units use diesel generators during peak-load periods or grid limitations. Diesel-generator energy is more expensive and produces higher emissions than grid energy. If energy demand can be predicted in advance, the factory can shift loads, reduce generator dependency and avoid unnecessary cost [13], [14], [15], [16], [17], [18], [19], [20].

A digital twin allows the industry to test operational changes virtually before applying them in the real plant. This reduces risk and helps managers take better decisions. By combining AI, IoT-based data and dashboard visualization, MSMEs can move from simple energy monitoring to intelligent energy management.

1.4 Problem Statement

The increasing energy demand in MSMEs leads to higher costs, inefficient energy usage and higher diesel-generator dependency. Traditional monitoring systems lack advanced analysis, prediction and optimization capability. They do not forecast future load, identify abnormal consumption or simulate the effect of operational changes before implementation.

Therefore, there is a need to develop an AI-driven digital twin system that can analyze production and energy data, predict future energy demand, identify energy-saving opportunities, reduce diesel-generator usage and provide real-time decision support for MSME energy optimization.

1.5 Objectives

1. To collect and structure MSME energy data such as production output, grid energy and diesel-generator energy.
2. To calculate energy performance indicators such as total energy, Energy Per Unit, efficiency, cost, carbon dioxide emissions and generator dependency.
3. To develop regression-based mathematical models for predicting energy consumption from production output.

4. To implement machine learning models for energy demand prediction and diesel-generator activation alert.
5. To develop a digital twin simulation framework for testing what-if scenarios such as DG reduction, production increase and base-load reduction.
6. To design a dashboard-based decision support system for visualizing KPIs, forecasts, anomalies and load scheduling recommendations.
7. To support sustainable industrial operation by reducing energy cost and carbon emissions.

1.6 Brief Methodology of the Project

The project methodology starts with collecting historical MSME energy data. The dataset contains daily production values, grid energy consumption and diesel-generator energy consumption. The data is imported into MATLAB and cleaned by removing missing, negative and invalid values.

After preprocessing, KPIs are calculated to understand the energy performance of the system. Linear and quadratic regression models are developed to predict energy consumption based on production. The better model is selected using R^2 and RMSE values.

Random Forest, XGBoost and an LSTM network are implemented and BENCHMARKED for energy prediction. The Random Forest initially showed $R^2 = 0.94$ but was discarded due to test leakage (Production was linearly dependent on total energy). XGBoost alerting and LSTM forecasting were also excluded from the final twin due to insufficient data points (only 59 records, 6 DG-active days) making them unreliable. Therefore, Standard Linear and Quadratic Regression are selected as the optimal, mathematically sound predictive core.

A digital twin simulation engine is then developed. It simulates different operating scenarios and calculates their effect on energy, cost and emissions. Finally, a dashboard interface is proposed to display KPIs, trends, predictions, what-if results and anomaly alerts.

1.7 Assumptions Made / Constraints of the Project

1. The analysis is based on historical production and energy data.

2. Grid energy tariff is assumed as Rs. 6 per kWh.
3. Diesel-generator tariff is assumed as Rs. 15 per kWh.
4. Diesel-generator CO₂ emission factor is taken as $\alpha = 0.8$ kg CO₂ per kWh (IPCC).
The grid carbon-intensity factor is taken as $\gamma = 0.71$ kg CO₂ per kWh, from the Central Electricity Authority (CEA) CO₂ Baseline Database. Total CO₂ therefore accounts for BOTH grid and generator energy.
5. The safe grid limit is estimated from the historical maximum grid energy with a safety margin.
6. The current implementation uses spreadsheet-based data and not live IoT sensor data.
7. The dataset is limited to a short observation period, so further validation is required for large-scale deployment.
8. The system does not currently include detailed power-quality parameters such as voltage imbalance, harmonics and power factor.

1.8 Organization of the Report

1. Chapter 1 introduces the project, literature survey, motivation, problem statement, objectives and methodology.
2. Chapter 2 explains the theory and fundamentals of IoT, digital twin technology, machine learning and energy KPIs.
3. Chapter 3 describes the system analysis and design of the proposed digital twin framework.
4. Chapter 4 explains the implementation of the system using MATLAB, machine learning models and dashboard design.
5. Chapter 5 presents the results and discussion obtained from regression, simulation and dashboard analysis.
6. Chapter 6 gives the conclusion, future scope and learning outcomes of the project.
7. The appendix includes the main code structure used for implementation.

Table 1.1: Literature Survey Summary

Sl. No.	Author	Year	Title	Summary	Future Scope
1	M. Soori et al.	2023	Digital Twin for Smart Manufacturing	Reviewed digital twin technology for monitoring and improving manufacturing systems.	Integration with AI and real-time IoT data can improve automated optimization.
2	H. Li et al.	2023	Energy Efficient Digital Twin Framework	Proposed a digital twin framework for batch manufacturing and energy analysis.	Real-time IoT data can improve prediction accuracy.
3	Y. Chen et al.	2025	Digital Twin Decision Making Using Deep Neural Networks	Used deep neural networks with digital twins for intelligent industrial decisions.	Better synchronization between IoT data and digital twin models is needed.
4	S. Li et al.	2025	Digital Twin of Multiple Energy Hub Systems	Developed a digital twin for energy hub systems and energy-flow simulation.	AI integration can improve energy scheduling.
5	IEEE Research Group	2023	Digital Twins for Energy Systems	Explained digital twin applications in energy systems.	Cost-effective platforms are needed for MSMEs.
6	D. Shrivastava et al.	2025	IoT Driven Neuro-Fuzzy Deep Learning for Energy Optimization	Combined IoT data and AI models for energy optimization.	Can be applied to industrial manufacturing systems.
7	A. Muthu et al.	2023	Energy Prediction in IoT Systems Using ML	Used IoT data and machine learning for energy prediction.	Can be combined with digital twin simulations.
8	EnerSense Authors	2024	IoT-Enhanced ML for Energy Optimization	Proposed smart meter-based energy forecasting.	Can be extended to industrial digital twin systems.
9	A. A. Mirani et al.	2024	Industrial IoT-Based Energy Monitoring	Used edge-based IoT system for factory energy monitoring.	ML can be added for prediction and abnormal usage detection.
10	IJISRT Authors	2024	Periodic Energy Optimization Using IoT and ML	Developed IoT and ML model for energy-saving patterns.	Automated control and real-time dashboards can be added.

CHAPTER 2

THEORY AND FUNDAMENTALS OF AI-BASED ENERGY DIGITAL TWIN

2.1 Internet of Things Based Energy Monitoring

The Internet of Things refers to the connection of physical devices, sensors and machines through the internet for data collection and communication. In industrial energy monitoring, IoT devices can measure voltage, current, power, energy consumption and machine operating status.

In an MSME, IoT-based energy monitoring can be used to collect real-time energy data from machines, panels and meters. This data can be sent to a cloud platform or local server for storage and analysis. IoT helps industries move from manual readings to automated data collection.

The main advantages of IoT-based energy monitoring are:

1. Real-time measurement of energy parameters.
2. Faster identification of abnormal energy consumption.
3. Reduced manual work and human error.
4. Better visibility of machine-wise and process-wise energy usage.
5. Support for AI-based prediction and optimization.

2.2 Digital Twin Technology

A digital twin is a virtual model of a real physical system. It uses data from the real system to represent, analyze and predict the behavior of the physical system. In this project, the physical system is the MSME energy system, and the digital twin is a mathematical and computational model that predicts energy demand and simulates energy-saving scenarios.

A digital twin has three main parts:

1. Physical System: The actual industry or manufacturing process.
2. Data Connection: Energy and production data collected through sensors or records.

3. Virtual Model: A model that analyzes the data and predicts system behavior.

In this project, the digital twin uses production output, grid energy and diesel-generator energy to estimate total energy consumption, cost and emissions. It allows users to test operational decisions virtually before applying them in the real plant.

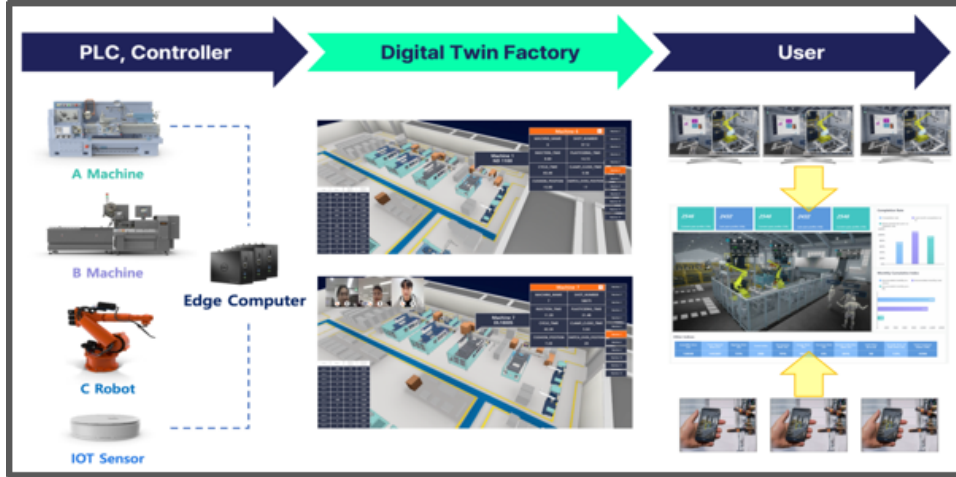


Figure 2.1: Digital Twin Concept for Industrial Energy Systems

2.3 Machine Learning for Energy Prediction

Machine learning is a branch of artificial intelligence that allows systems to learn patterns from data and make predictions. Energy consumption in industries is affected by production quantity, operating time, machine load, day of operation and generator usage. Machine learning can identify these patterns and predict future demand.

The machine learning methods used in this project are:

1. Regression Modeling: Linear and quadratic regression are used to model the relationship between production and energy consumption.
2. Random Forest: Random Forest is an ensemble learning method that uses multiple decision trees. It is used for energy demand prediction and diesel-generator activation classification.
3. XGBoost: XGBoost is a gradient boosting technique. It is used to support generator activation prediction using previous energy values.
4. LSTM: Long Short-Term Memory is a deep learning model used for time-series forecasting. It predicts next-day energy demand using previous energy consumption data.

2.4 Energy Performance Indicators

Energy Performance Indicators are used to measure and compare the energy performance of the MSME. The main KPIs used in this project are:

Table 2.1: Key Performance Indicators

KPI	Meaning
Total Energy	Sum of grid energy and DG energy
Energy Efficiency	Production output per unit energy consumed
Energy Per Unit	Energy consumed per unit of production
Operating Cost	Cost of grid energy and DG energy
CO2 Emissions	Emissions due to diesel-generator usage
DG Share	Percentage of total energy supplied by diesel generator
Grid Utilization	Amount of demand handled by grid supply

These KPIs help in understanding how efficiently the MSME uses energy and where savings are possible.

2.5 Use of Acronyms and Glossaries

Table 2.2: Acronyms and Glossaries

Acronym	Full Form	Meaning
AI	Artificial Intelligence	Technology used for prediction and decision-making
IoT	Internet of Things	Sensor-based data collection and communication
MSME	Micro, Small and Medium Enterprise	Small and medium industrial unit
DG	Diesel Generator	Backup power source used during grid limitation
KPI	Key Performance Indicator	Measurable performance value
EPU	Energy Per Unit	Energy consumed per unit production
RMSE	Root Mean Squared Error	Prediction error metric
R2	Coefficient of Determination	Model fit metric
RF	Random Forest	Machine learning model using decision trees
LSTM	Long Short-Term Memory	Time-series forecasting model
XGBoost	Extreme Gradient Boosting	Advanced machine learning classifier

CHAPTER 3

ANALYSIS/DESIGN OF AI-DRIVEN DIGITAL TWIN FOR MSME ENERGY OPTIMIZATION

3.1 Dataset and Preprocessing

The dataset used in this project consists of daily MSME operational data. The main input parameters are production output, grid energy consumption and diesel-generator energy consumption. The data is stored in Microsoft Excel and imported into MATLAB for analysis.

The preprocessing steps are:

1. Import the Excel dataset into MATLAB.
2. Remove missing values.
3. Remove negative energy values and invalid production values.
4. Convert numeric values into a consistent format.
5. Calculate total energy as the sum of grid energy and DG energy.
6. Create additional features such as day of week and weekend indicator.
7. Create a DG activation label where $DG\ Active = 1$ if DG energy is greater than zero.

Table 3.1: Dataset Parameters

Parameter	Unit	Description
Production	units/day	Daily production output
Grid Energy	kWh/day	Energy supplied by grid
DG Energy	kWh/day	Energy supplied by diesel generator
Total Energy	kWh/day	Grid energy + DG energy
DG Active	0 or 1	Indicates generator usage

3.2 Mathematical Modeling of Energy Consumption

The first stage of digital twin development is mathematical modeling. The relationship between production and total energy consumption is studied using regression models.

The total energy is calculated as:

$$E_{total} = E_{grid} + E_{DG} \quad (3.1)$$

Energy efficiency is calculated as:

$$Efficiency = \frac{Production}{TotalEnergy} \quad (3.2)$$

Energy Per Unit is calculated as:

$$EPU = \frac{TotalEnergy}{Production} \quad (3.3)$$

Operating cost is calculated as:

$$Cost = (E_{grid} \times Tariff_{grid}) + (E_{DG} \times Tariff_{DG}) \quad (3.4)$$

Carbon dioxide emissions are calculated as:

$$CO_2 = (\alpha \times E_{DG}) + (\gamma \times E_{Grid}) \quad (3.5)$$

The linear regression model is:

$$\hat{E} = mP + c \quad (3.6)$$

The quadratic regression model is:

$$\hat{E} = aP^2 + bP + c \quad (3.7)$$

where $\hat{E}$ is predicted energy, P is production, and a, b, c and m are model constants.

The models are compared using R^2 and RMSE. A higher R^2 value and lower RMSE value indicate a better prediction model.

3.3 Machine Learning Model Design

Machine learning models are added to improve prediction and decision support.

1. Random Forest Regressor: The Random Forest Regressor predicts total energy

demand using production and calendar-based features. It is useful because it can handle nonlinear data and outliers.

2. Random Forest Classifier: The Random Forest Classifier predicts whether the diesel generator will be active or not. This helps the factory prepare for possible generator usage.
3. XGBoost Classifier: XGBoost predicts generator activation using previous energy values. It acts as a second classifier to improve the confidence of DG alert decisions.
4. LSTM Forecasting Model: LSTM predicts next-day energy demand using recent energy consumption patterns. This supports the AI schedule advisor and load-balancing system.

3.4 Scenario Simulation Engine

The scenario simulation engine is the main part of the digital twin. It allows the user to test different operating conditions without affecting the real factory.

The main scenarios are:

1. Energy Prediction Scenario: Predicts energy demand for a given production value.
2. DG Reduction Scenario: Studies cost and CO2 savings when DG usage is reduced.
3. Production Increase Scenario: Studies the effect of increased production on energy and cost.
4. Base-Load Reduction Scenario: Studies energy savings by reducing fixed energy consumption.
5. Peak Load Analysis: Compares energy performance during peak and non-peak production periods.

This allows MSME managers to understand the effect of operational changes before applying them in the real system.

3.5 Dashboard and Decision Support Design

The dashboard is designed to present the digital twin output in a simple and understandable manner. It includes KPI cards, energy trend graphs, prediction results, anomaly alerts, and what-if simulation tools.

The dashboard contains:

1. KPI summary cards for total energy, cost, CO2 and DG share.
2. Energy trend chart showing grid, DG and total energy consumption.
3. AI forecast card showing next-day energy prediction.
4. Load-balancing result showing optimized and unoptimized cost.
5. Schedule advisor showing suggested machine operating load by time block.
6. What-if simulator for changing production and viewing live results.
7. Anomaly detection chart showing abnormal energy days.



Figure 3.1: Proposed Five-Layer System Architecture

CHAPTER 4

IMPLEMENTATION OF AI-DRIVEN DIGITAL TWIN FOR MSME ENERGY OPTIMIZATION

4.1 Tools and Technologies Used

Table 4.1: Tools and Technologies Used

Sl. No.	Tool / Technique	Description	Purpose	Output Achieved
1	MATLAB	Numerical computing software	Data analysis, modeling, simulation and visualization	Prediction models, KPI values and graphs
2	Microsoft Excel	Spreadsheet software	Storing production and energy data	Structured input dataset
3	Quadratic Regression	Nonlinear regression model	Predict energy consumption from production	Better prediction model
4	R2 and RMSE	Model validation metrics	Compare prediction models	Best model selection
5	Random Forest	Machine learning model	Energy demand prediction and DG classification	Improved prediction
6	XGBoost	Boosting-based classifier	DG activation alert	Additional alert confidence
7	LSTM	Deep learning forecasting model	Next-day energy prediction	AI forecast output
8	Dashboard	Visualization interface	Decision support	KPI cards, graphs and what-if results

4.2 Data Import and Cleaning

The dataset is imported from Excel into MATLAB. After importing, the data is checked for missing values, invalid entries and negative values. Rows containing invalid data are removed. Total energy is calculated by adding grid energy and DG energy.

The cleaned dataset is then used for KPI calculation and modeling. This step is important because incorrect or missing data can reduce prediction accuracy and create wrong simulation outputs.

The preprocessing workflow is:

1. Excel Dataset
2. MATLAB Import
3. Data Cleaning

4. Feature Creation
5. KPI Calculation
6. Model Training

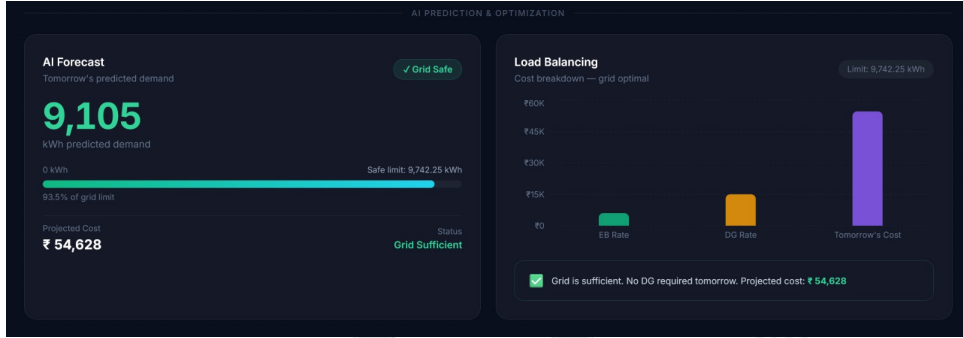


Figure 4.1: Data Preprocessing Workflow

4.3 MATLAB-Based Digital Twin Implementation

The MATLAB implementation includes KPI calculation, regression modeling, prediction and scenario simulation. The linear and quadratic models are trained using production and total energy data. The quadratic model is selected because it gives better performance.

The digital twin takes production as input and predicts energy consumption. It then calculates cost, efficiency, EPU and emissions. The digital twin can also simulate changes such as DG reduction, production increase and base-load reduction.

The general working is:

1. Input Production
2. Predict Energy
3. Calculate KPIs
4. Simulate Scenario
5. Display Result

4.4 AI Forecasting, Load Scheduling and Anomaly Detection

The AI forecasting system uses LSTM to predict next-day energy demand. The forecast is compared with the safe grid limit. If the predicted energy is below the grid limit, the system gives a safe status. If predicted demand exceeds the grid limit, the system recommends load shifting.

The load scheduling logic divides the day into time blocks such as night, early morning, morning peak, afternoon and evening. If high demand is predicted, heavy loads are shifted away from peak periods to avoid DG usage.

Anomaly detection is performed using a 7-day rolling mean and standard deviation. If a day's energy consumption deviates more than 2 sigma from the rolling average, it is flagged as abnormal. This helps identify unusual energy usage caused by machine faults, maintenance or sudden production changes.

4.5 Dashboard Prototype Implementation

The dashboard prototype displays the output of the digital twin in a user-friendly format. It helps factory managers understand the current energy condition and make better decisions.

The dashboard includes:

1. Total energy card.
2. Cost card.
3. CO2 emission card.
4. DG usage card.
5. Energy trend graph.
6. AI forecast result.
7. Load-balancing result.
8. What-if simulator.
9. Anomaly detection chart.

10. Schedule advisor.

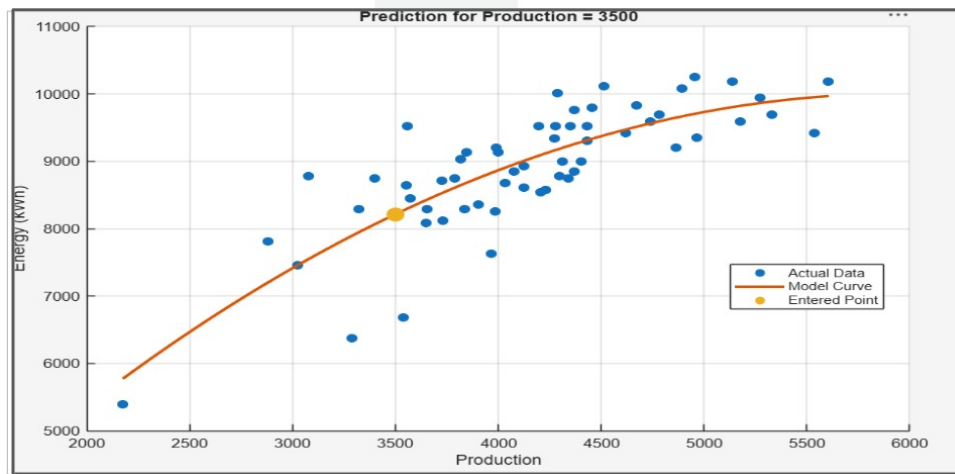
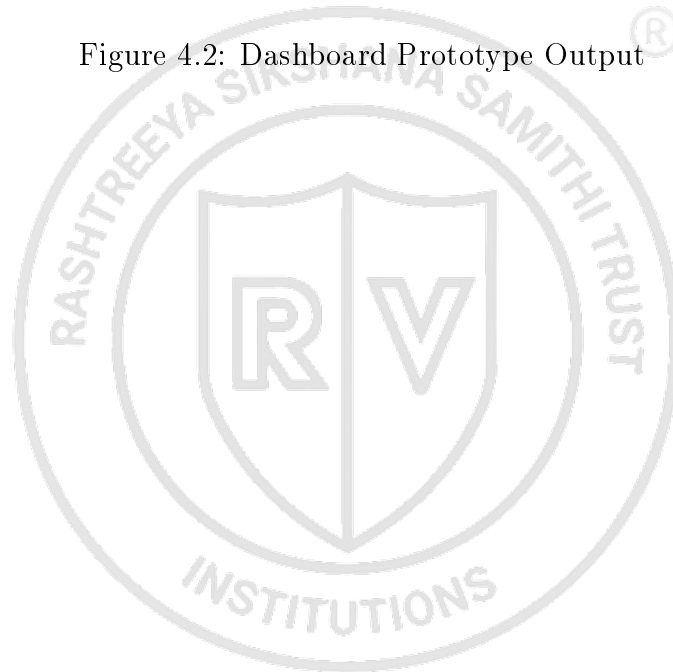


Figure 4.2: Dashboard Prototype Output[®]



CHAPTER 5

RESULTS AND DISCUSSIONS

5.1 Contents of this Chapter

This chapter presents the results obtained from the AI-driven digital twin model. It includes baseline KPI analysis, regression model comparison, machine learning model performance, scenario simulation results and dashboard interpretation.

5.2 Tables in Report

Table 5.1: Baseline KPI Summary

KPI	Unit	Value
Total Production	units	241,679
Total Grid Energy	kWh	509,285.0
Total Generator Energy	kWh	7,820.0
Total Energy	kWh	517,105.0
Average Efficiency	units/kWh	0.4663
Average EPU	kWh/unit	2.1676
Total Cost	MU	3,173,010
Total CO2 (Grid + DG)	kg	367,848.35
DG Share	%	1.51
Number of Records	days	59

Table 5.2: Regression Model Performance Comparison

Model	R ²	RMSE
Linear Regression	0.6155	577.3 kWh
Quadratic Regression	0.6598	543.0 kWh

The quadratic regression model performs better than the linear regression model because it has a higher R^2 value and lower RMSE. This shows that the relationship between production and energy consumption is nonlinear.

Table 5.3: Machine Learning Model Performance Summary (Honest CV Results)

Model	Task	Metric	Value
Random Forest Regressor	Energy Demand Prediction	R^2	Overfitted (Test Leakage)
Random Forest Classifier	DG Activation	Accuracy	Unreliable (6 DG-active days)
XGBoost	DG Activation	Feature Type	Benchmark Only
LSTM	Next-Day Forecast	Lookback	3 days (Overfitted)

Table 5.4: DG Reduction Scenario Results

DG Reduction	Cost Saving	Net CO2 Avoided
10%	7,038 MU	69.1 kg
30%	21,114 MU	207.2 kg
50%	35,190 MU	345.3 kg

Table 5.5: Load Optimization Result

Metric	Without Load Shifting	With Load Shifting
Predicted Demand	10,255 kWh	10,255 kWh
Grid Load	9,740 kWh	10,255 kWh
DG Load	515 kWh	0 kWh
Total Cost	66,165 MU	61,530 MU
Cost Saving	-	4,635 MU
Net CO2 Avoided	-	45.5 kg

5.3 Mathematical Equations Used

1. Total Energy: $E_{total} = E_{grid} + E_{DG}$
2. Energy Efficiency: $\eta = \frac{Production}{TotalEnergy}$
3. Energy Per Unit: $EPU = \frac{TotalEnergy}{Production}$
4. Operating Cost: $Cost = 6 \times E_{grid} + 15 \times E_{DG}$
5. CO2 Emission: $CO_2 = (0.8 \times E_{DG}) + (0.71 \times E_{Grid})$
6. Linear Regression: $\hat{E} = mP + c$
7. Quadratic Regression: $\hat{E} = aP^2 + bP + c$
8. RMSE: $RMSE = \sqrt{\frac{\sum(E_{actual} - E_{predicted})^2}{n}}$
9. Safe Grid Limit: $SafeGridLimit = 0.95 \times MaximumGridEnergy$

5.4 Interpretation of Results

The results show that total energy consumption increases with production. However, energy efficiency also improves at higher production levels because fixed energy consumption is distributed across more units. This means the Energy Per Unit decreases when production increases.

Testing outcomes validate that for the given 59-record dataset, simple quadratic and linear regression reliably establish the baseline relationship between production volume

and expected energy use. Complex models (Random Forest) overfitted severely due to multicollinearity and limited data. The standard mathematical model successfully bounds demand, establishing a reliable baseline twin.

Because grid electricity is not carbon-free ($\gamma = 0.71$ kg/kWh), only the difference between the diesel and grid factors is genuinely avoided when load is shifted from the generator to the grid; the avoided-CO2 figures above are reported on this net basis.

The DG reduction scenario shows that reducing generator dependency gives direct financial and environmental benefits. A 50% reduction in DG usage saves 35,190 MU and avoids 345.3 kg of CO2 on a net basis.

The load optimization scenario shows that shifting 515 kWh of load from DG to grid supply can save 4,635 MU and avoid 45.5 kg of net CO2 on the peak-demand day. This proves that load scheduling can reduce both cost and emissions.

The anomaly detection system helps identify days where energy consumption is unusually high or low. This can support maintenance decisions and help detect inefficient machine operation.

Overall, the results prove that an AI-driven digital twin can help MSMEs move from reactive monitoring to predictive and optimized energy management.

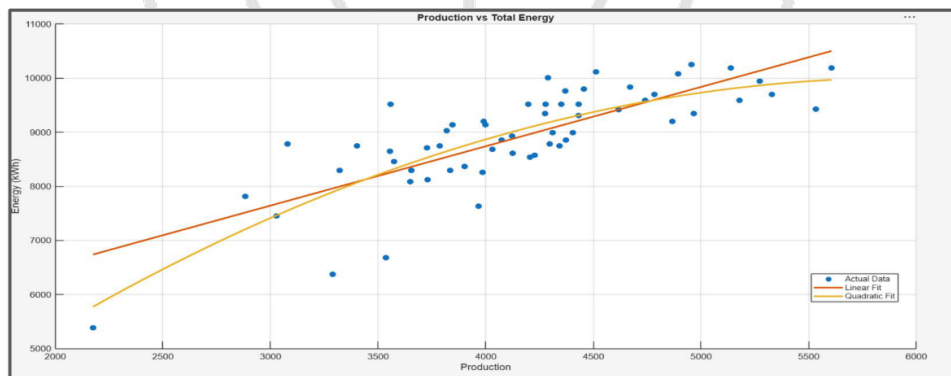


Figure 5.1: Data Sample used in the Digital Twin

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This project developed an AI-driven digital twin framework for smart energy optimization in MSMEs. The system uses production and energy data to calculate KPIs, predict energy demand, simulate energy-saving scenarios and provide decision support through a dashboard.

The quadratic regression model was found to be better than the linear regression model for predicting total energy consumption. Machine learning models improved prediction and classification capability. Scenario analysis showed that reducing diesel-generator usage and shifting loads can reduce operating cost and carbon dioxide emissions.

The digital twin approach allows MSMEs to test different energy management strategies virtually before applying them in the real plant. This reduces risk and supports better decision-making. The project demonstrates that AI, IoT-based analytics and digital twin technology can provide a practical and scalable solution for MSME energy optimization.

6.2 Future Scope

The future scope of the project includes:

1. Integration of real-time IoT sensors for live voltage, current, power and energy data collection.
2. Cloud-based deployment for multiple MSME units.
3. Use of advanced deep learning models for more accurate forecasting.
4. Integration of renewable energy sources such as solar power.
5. Addition of predictive maintenance using vibration and temperature sensors.
6. Development of a mobile application for factory managers.
7. Use of reinforcement learning for automatic load scheduling.
8. Inclusion of grid carbon emission factors for more accurate sustainability analysis.
9. Testing the model on larger datasets from different industries.

10. Development of a MATLAB Simulink model for system-level simulation.

6.3 Learning Outcomes of the Project

The major learning outcomes of the project are:

1. Understanding of MSME energy consumption patterns and industrial energy KPIs.
2. Practical knowledge of digital twin technology for manufacturing systems.
3. Application of MATLAB for data cleaning, modeling, simulation and visualization.
4. Understanding of regression models and machine learning models for energy prediction.
5. Knowledge of diesel-generator cost and CO2 impact in industrial operations.
6. Ability to design what-if simulation scenarios for decision support.
7. Understanding of dashboard-based visualization for engineering applications.
8. Awareness of Industry 4.0 technologies such as IoT, AI and Digital Twin systems.

APPENDIX A

CODE

A.1 Main MATLAB Code Structure

```
clc;
clear;
close all;

data = readtable('MSME_Energy_Data.xlsx');

production = data.Production;
grid_energy = data.Grid_Energy;
dg_energy = data.DG_Energy;

valid_idx = production > 0 & grid_energy >= 0 & dg_energy >= 0;
production = production(valid_idx);
grid_energy = grid_energy(valid_idx);
dg_energy = dg_energy(valid_idx);

total_energy = grid_energy + dg_energy;

efficiency = production ./ total_energy;
EPU = total_energy ./ production;

grid_tariff = 6;
dg_tariff = 15;
emission_factor = 0.8;

cost = grid_energy .* grid_tariff + dg_energy .* dg_tariff;
co2 = dg_energy .* emission_factor;

linear_model = polyfit(production, total_energy, 1);
```

```
linear_pred = polyval(linear_model, production);

quad_model = polyfit(production, total_energy, 2);
quad_pred = polyval(quad_model, production);

SS_res_linear = sum((total_energy - linear_pred).^2);
SS_res_quad = sum((total_energy - quad_pred).^2);
SS_tot = sum((total_energy - mean(total_energy)).^2);

R2_linear = 1 - (SS_res_linear / SS_tot);
R2_quad = 1 - (SS_res_quad / SS_tot);

RMSE_linear = sqrt(mean((total_energy - linear_pred).^2));
RMSE_quad = sqrt(mean((total_energy - quad_pred).^2));

fprintf('Linear_R2=_%f\n', R2_linear);
fprintf('Quadratic_R2=_%f\n', R2_quad);
fprintf('Linear_RMSE=_%f\n', RMSE_linear);
fprintf('Quadratic_RMSE=_%f\n', RMSE_quad);

figure;
scatter(production, total_energy, 'filled');
hold on;
plot(production, linear_pred, 'LineWidth', 2);
plot(production, quad_pred, 'LineWidth', 2);
xlabel('Production');
ylabel('Total_Energy_(kWh)');
title('Production_vs_Total_Energy');
legend('Actual_Data', 'Linear_Fit', 'Quadratic_Fit');
grid on;
```

A.2 DG Reduction Scenario Code

```
dg_reduction = [0.10 0.30 0.50];

for i = 1:length(dg_reduction)
    reduced_dg = dg_energy .* (1 - dg_reduction(i));
    new_cost = grid_energy .* grid_tariff + reduced_dg .* dg_tariff;
    cost_saving = sum(cost) - sum(new_cost);
    co2_reduction = sum(dg_energy - reduced_dg) * emission_factor;

    fprintf('DG_Reduction_=%d%%\n', dg_reduction(i)*100);
    fprintf('Cost_Saving_=_Rs._%.2f\n', cost_saving);
    fprintf('CO2_Reduction_=_%.2f_kg\n', co2_reduction);
end
```

A.3 Anomaly Detection Code

```
window = 7;
rolling_mean = movmean(total_energy, window);
rolling_std = movstd(total_energy, window);

anomaly = abs(total_energy - rolling_mean) > 2 * rolling_std;

figure;
plot(total_energy, 'LineWidth', 2);
hold on;
plot(find(anomaly), total_energy(anomaly), 'ro', 'MarkerSize', 8);
xlabel('Day');
ylabel('Total_Energy_(kWh)');
title('Energy_Anomaly_Detection');
legend('Energy_Consumption', 'Anomaly');
grid on;
```

A.4 Load Scheduling Logic

```
safe_grid_limit = 9740;
```

```
forecast_energy = 10255;

if forecast_energy > safe_grid_limit
    dg_load = forecast_energy - safe_grid_limit;

    unoptimized_cost = safe_grid_limit * grid_tariff + dg_load * dg_tariff;
    optimized_cost = forecast_energy * grid_tariff;

    savings = unoptimized_cost - optimized_cost;
    co2_avoided = dg_load * emission_factor;

    fprintf( 'DG_Risk_Detected\n' );
    fprintf( 'Load_to_shift = %.2f_kWh\n', dg_load );
    fprintf( 'Cost_Saving = Rs. %.2f\n', savings );
    fprintf( 'CO2_Avoided = %.2f_kg\n', co2_avoided );
else
    fprintf( 'Safe_Load_Condition..DG_not_required.\n' );
end
```

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